PROJECT CONCEPT NOTE: INIDA-ML

Intelligent Network Intrusion Detection and Anomaly Analysis Using Machine Learning

Administrative & Institutional Profile

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1. Problem Statement & Motivation

Modern enterprise networks and connected digital infrastructures generate massive volumes of tabular flow telemetry consisting of both benign user traffic and malicious cyber incursions. Manual packet inspection and signature maintenance are slow, reactive, and prone to evasion. This project establishes an intelligent machine learning workflow to ingest network flow characteristics, classify anomalies efficiently, and display actionable intelligence through a structured web application workbench.

2. UN Sustainable Development Goal (SDG) Alignment

SDG 9: Industry, Innovation, and Infrastructure

INIDA-ML directly supports building resilient digital infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation by providing accessible, open-source automated intrusion detection tools for academic research and small IT environments.

3. Methodology & Core Objectives

- **Data Validation & Ingestion:** Validate records against the CIC-IDS2017 77-feature schema, screening missing values and preventing target leakage.
- **Supervised & Unsupervised Models:** Train a resource-conscious Random Forest classifier (and XGBoost/Autoencoders where attack data is present). Note: The initial `Benign-Monday-no-metadata.parquet` file (458,831 records) strictly contains benign traffic and cannot train binary classifiers alone.
- **Web Application & Auxiliary AI:** Build an interactive Streamlit workbench featuring traffic overviews, CSV analysis, live port probing, URL threat inspection, and optional Groq AI explanations.

Customer Segments

- Students & Educators learning cybersecurity
- Researchers evaluating ML classifiers
- Small IT teams testing flow prototypes

Key Metrics & Validation

- Precision, Recall, Macro-F1, Balanced Accuracy
- Confusion matrix and false-positive rates
- Pytest coverage & JSON history tracking

4. Technology Stack & Deliverables

Stack: Python, Streamlit, pandas, NumPy, scikit-learn (Random Forest, joblib), Plotly visualizations, Groq API (optional), and pytest.

Project Deliverables: Fully functional Streamlit web application repository on GitHub, accompanying PowerPoint presentation deck, and this formal Concept Note document.

5. Scope, Limitations & Ethical Disclaimer

Limitations: Monday-only benign data is insufficient for binary training without attack captures. Cloud deployments cannot capture a visitor's private network directly. AI explanations from Groq,Gemini LLM are supplementary and do not replace deterministic model outputs or expert human security oversight.

6. Project Lean Canvas

Structured strategic overview validating business assumptions, target audience, and architectural mechanics for INIDA-ML.

1. PROBLEM

- Large volumes of flow records overwhelm manual review.
- Benchmark models produce false positives or fail to generalize.
- Alerts & history spread across disconnected tools.

2. SOLUTION

- Random Forest & XGBoost on CIC-IDS2017 flows.
- Autoencoder anomaly scoring.
- Local live-flow monitoring & offline CSV analysis.

3. KEY METRICS

- Macro-F1, precision, recall, FPR.
- Throughput & inference latency.
- Resource utilization.

4. VALUE PROP

A local-first, explainable learning prototype combining supervised classification, anomaly scoring, live-flow integration, alerts, and reports in a unified dashboard.

5. UNFAIR ADV.

- Integrated ML + DL workflow.
- Transparent local evaluation.
- Student-friendly modular codebase.

6. SEGMENTS

- Cybersecurity students & educators.
- ML researchers.
- Small IT audit teams.
- Defensive developers.

7. ALTERNATIVES

- Rule-based IDS/IPS.
- Firewall logs & manual packet inspection.
- Commercial SIEM tools.

8. CHANNELS

- GitHub repository.
- Streamlit Cloud demo.
- Academic presentations.

9. EARLY ADOPTERS

- College security labs.
- ML project evaluators.
- Authorized testers.

10. COST STRUCTURE

Development time, laptop compute, dataset processing, automated testing, optional hosting, and maintenance.

11. REVENUE

Academic & portfolio baseline. Potential future options: support services, hosted analytics, workshops. (No revenue assumed currently).

Concept Validation Note: Designed as an academic prototype under the IBM SkillsBuild & AICTE internship program. Relevance to UN SDG 9 is maintained through open-source accessibility.